

we will talk about communication between cells, so we're going to talk about multi-cellular organisms (unicellular). you need to respond signals, otherwise you will not survive. So a lot of times every second the cells respond to signals. 5/11/2018

D

CHAPTER 15

Cell Signaling and Signal Transduction: Communication Between Cells



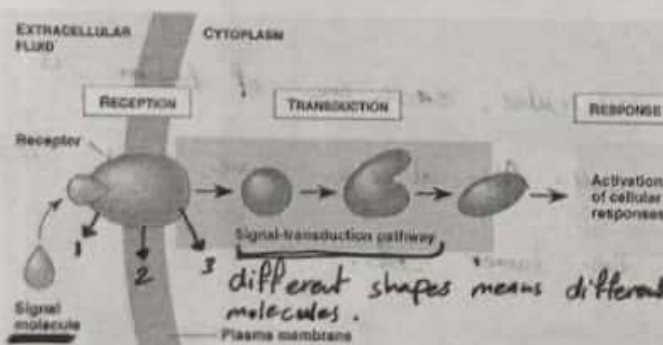
- you have to be equipped with the mechanism to respond to the signal. which a lot of the times you will be thinking of cell surface receptors. but we will talk about that not only cell surface receptors, also other things the cell have, so that we have respond to the signal.

Introduction

- Cells must respond adequately to external stimuli to survive.
- Cells respond to stimuli via cell signaling.
- Some signal molecules enter cells; others bind to cell-surface receptors.

A signal molecule binds to the receptor, by definition the Receptor :- 5, 1, 2, 3
 (if it's a cell surface receptor it has 3 domains :- 1) outside → extracellular binding domain, which means that it's ~~designed~~ designed to fit the signal molecule. 5/1/2018
 2) in the plasma membrane there is a transmembrane domain.
 3) cytoplasmic domain: is basically the idea of signal transduction.

General mechanism of cellular signaling



anything the cell responds to
 from amino acids to
 ions, proteins, growth factor ...

1. Reception of signal
2. Transduction of signal
3. Cellular response

2) - the signal molecule itself will not go inside.

what goes inside is the result of binding of the signal to Receptor.

so what happens to the Receptor → the change in conformation through the cytoplasmic domain.

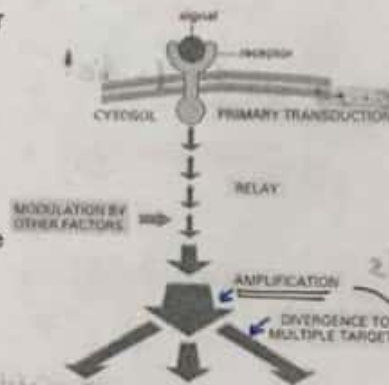
3) later we will get to know that

1. Reception of signal
2. Transduction of signal
3. Cellular response

Receptor → the change in conformation through the cytoplasmic domain.

Signaling cascades perform 5 crucial functions

1. Transduce signal into molecular form that can stimulate response
- Relay signal from point of reception to point of action in the cell
- Amplify the received signal
- Distribute the signal to influence several responses in parallel
- Each step is open to modulation by other signals



3) later we will get to know that one of these domains what we call it "a cytoplasmic Kinase domain"

It means that the domain will be phosphorylated.

amino acids

all group

→ the cytoplasmic domain will be phosphorylated. So

it can initiate, or stop the pathway.

2

Amplification:- there was one signal. you need to amplify the signal.

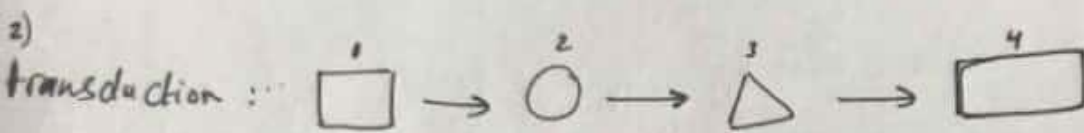
And to do that within the cell there's a system → "second messenger"

For example :- cyclic AMP . but at the same time later there will be divergence for the signal (to go to different places)

Now:- the arrows are 3. which means :- the binding to 1st molecule causes 3 different processes.

1) Reception:- the signal binding to the Receptor
when there's No Reception in the cell.
what's the problem??

Receptor site



A multistep process that happens through the transduction.

3) Response:- which is basically the function. If you're binding to a growth factor that means the cell wants to ^{grow} divide to: ^{1) grow} ^{2) Repair}.
So the Response to this signal will be to get into the cell cycle so the cell divide.

proteins have shapes through its folding.

- the original signal (primary signal).
primary messenger

Response:- which is basically the function of the signal factor. that means the cell wants to ^{1) grow} divide to: ^{2) Repair} So the Response to this signal will be to get into the cell cycle so the cell divide.

proteins have shapes through its folding.

- the original signal (primary signal).
primary messenger

And sometimes there's a secondary messenger.

secondary transduction the signal primary transduction

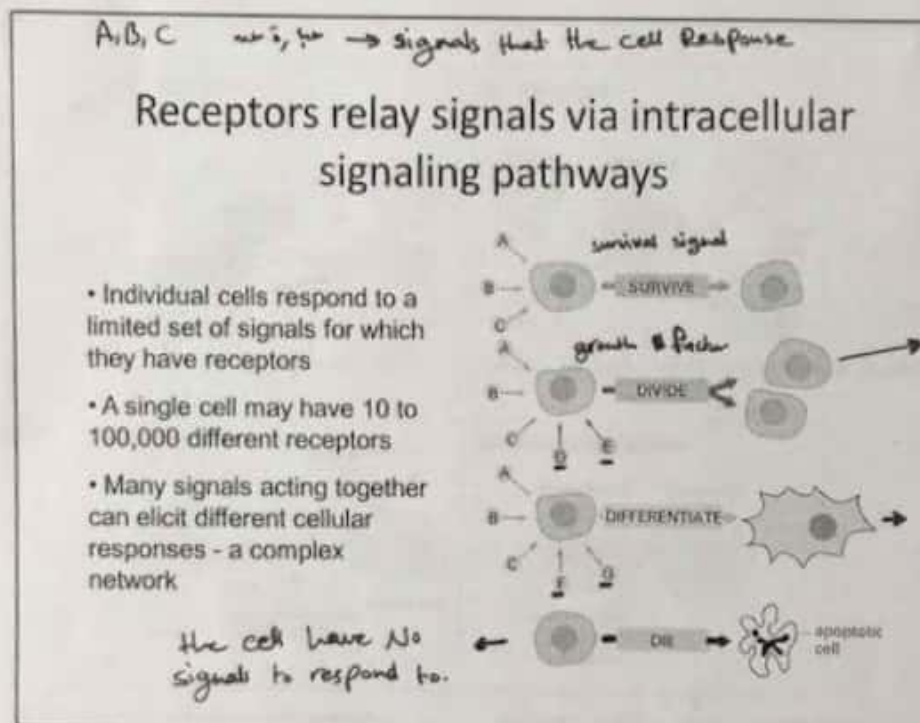
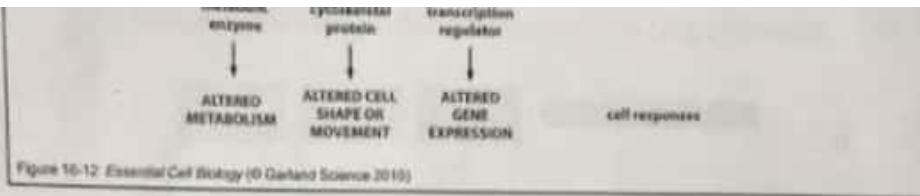
Relay:- one gives to another
Is the idea of a signal pathway is to do something, then that one thing is interfering with the other thing to happen

pathway in the signal pathway.

Relay :- steps 1 → 2 → 3 → 4 → 5
before you have 4, you need 3

upstream:- the previous steps

downstream:- the steps that will happen



Cell have choices:-
 1) survive
 2) divide
 3) differentiate
 4) Die

Cells divide for a reason.

Cells normally, they don't want to keep dividing. 99% are not dividing.

they are already differentiated → they are done dividing → they do specific function now → they have specific structure, function.

Normal cells need a growth factor to divide.

self sufficiency of Growth signals

hallmarks of cancer

Ok cells: tumor cells / bi. Lolo

they make the growth factors inside. then secrete it outside. the cells think that they are responding to growth factors.

Modulation:- Multi steps of Regulation.

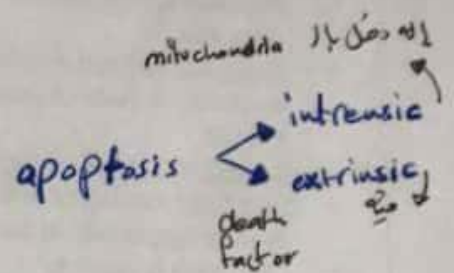
* it's Not one steps, and one enzyme! It's multi different steps
So that anytime there is a problem in one, things can be
fixed by another.

- This good for us and for the tumor.

Modulation is

these different signaling molecules, each one of them is
a target for the tumor cell. Also at the same time it's a target
for the drug to target the tumor cells.

- Cancer cells don't want to die!
(Hallmarks of cancer:-
evasion of apoptosis.
they inactivate apoptosis



If you're a Normal cell and you have a problem
you better die.

P53: is called "Guardian of the genome"

↓
tumor suppressor protein

• can be mutated, 1/3!

50% of the tumors have mutant p53.

Signaling Systems (1)

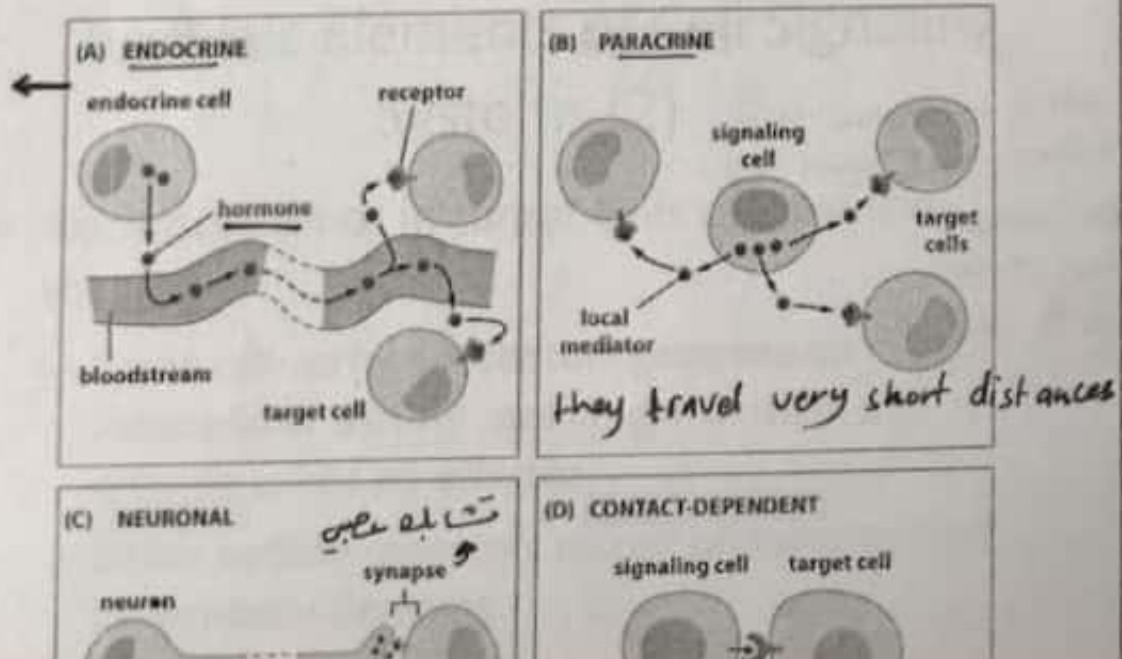
- Extracellular messenger molecules transmit messages between cells.

- In **autocrine** signaling, the cell has receptors on its surface that respond to the messenger.
- During **paracrine** signaling, messenger molecules travel short distances through extracellular space.
- During **endocrine** signaling, messenger molecules reach their target cells through the bloodstream.

is found
the signal within
the cell, secreted
outside. and the
response process is
happening on the
same cell.

like cancer cells.
they don't rely
on other cells,
they do this

insulin are secreted
from beta islets
of the pancreas,
traveling through
the blood stream
to go activate
different types of cells.



they do this

insulin are secreted from beta islets of the pancreas, traveling through the blood stream to go activate different types of cells.

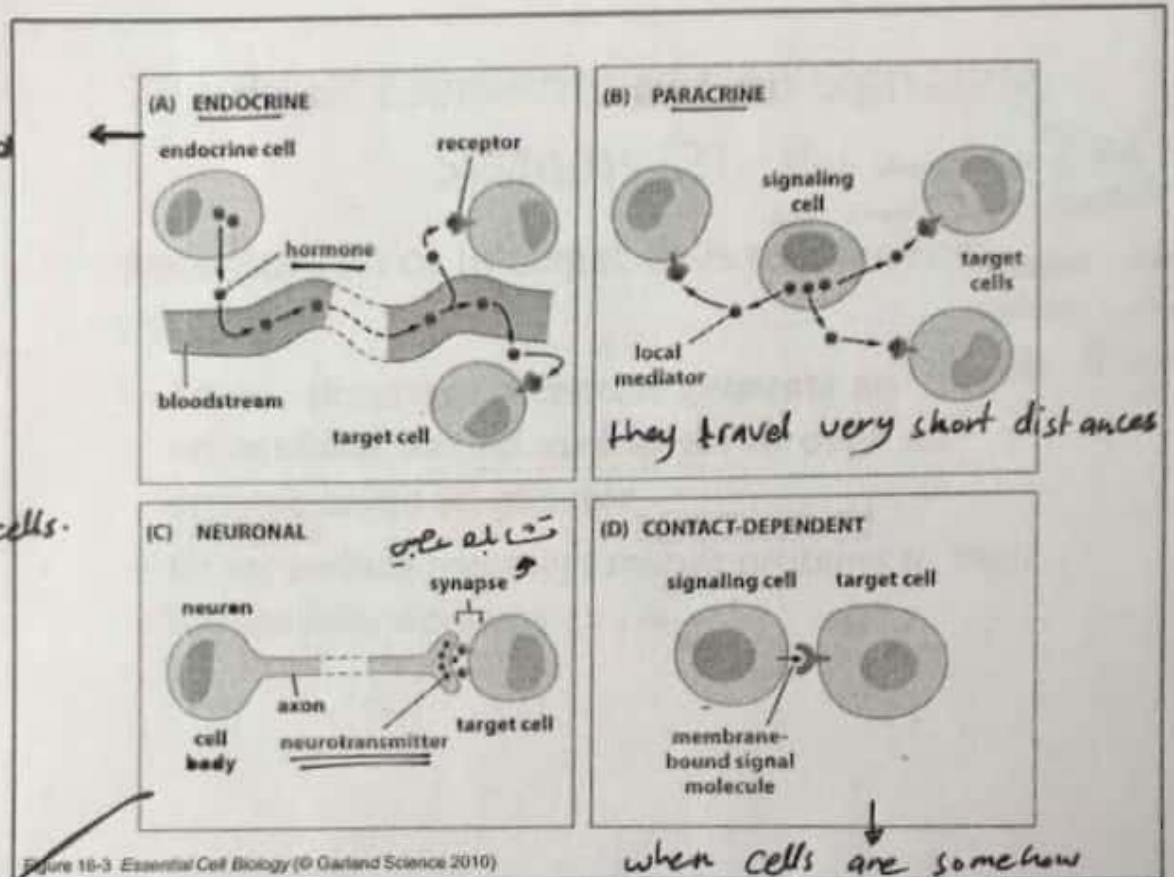


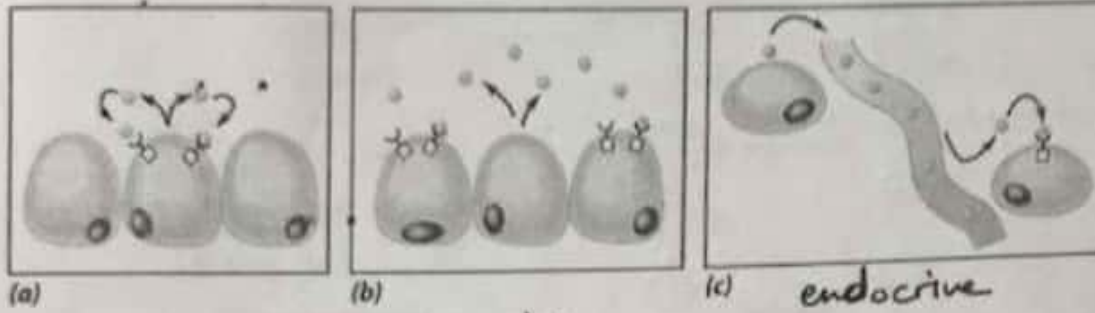
Figure 16-3 Essential Cell Biology (© Garland Science 2010)

axons are very long

The signal travel through the axons to reach the dendrites, area we call the synapse, then neurotransmitters leave ~~these~~ through these synapse (synaptic transmission) to affect a target.

when cells are somehow bound to each other. they send signals to contact. "happens alot within tissues"

Types of intercellular signaling



(a) autocrine

(b) paracrine

(c) endocrine

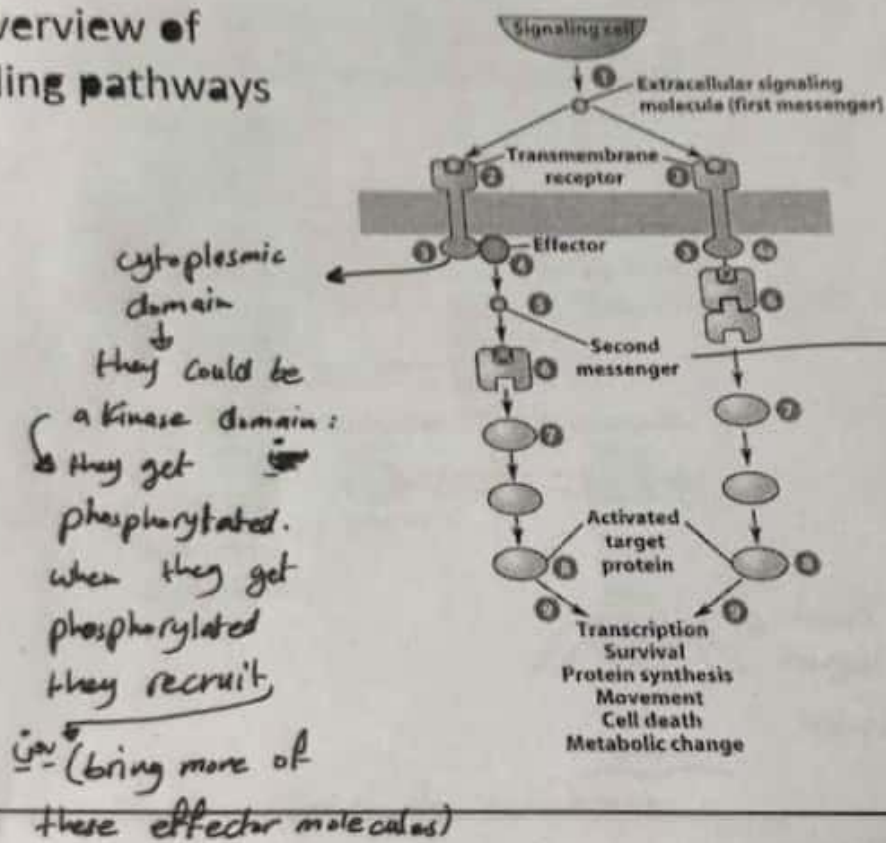
The Basic Elements of Cell Signaling

Systems (2)

- Receptors on or in target cells receive the message.
 - Some cell surface **receptors** generate an intracellular **second messenger** through an enzyme called an **effector**.

First messenger (the signal) but they don't call it ~~that~~ First messenger because second messengers are limited, what they are. while signals are endless. *202*

● Overview of signaling pathways

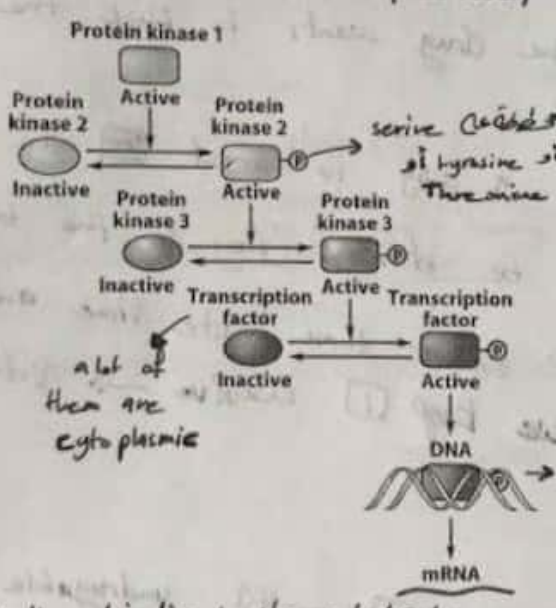


when the second messenger is involved, there will be amplification

1) Kinase: an enzyme capable of taking the phosphate, putting a different phosphate somewhere else:
 the source of phosphate could be ATP and could be from "a substrate that has phosphate."
 - the most important modification that could happen to a protein is phosphorylation. 5/1/2018
 phosphorylation can target 3 amino acid: tyrosine, Threonine, serine

what's common between them? they have a hydroxyl group that can some how bind to the phosphate group.

A signal transduction pathway

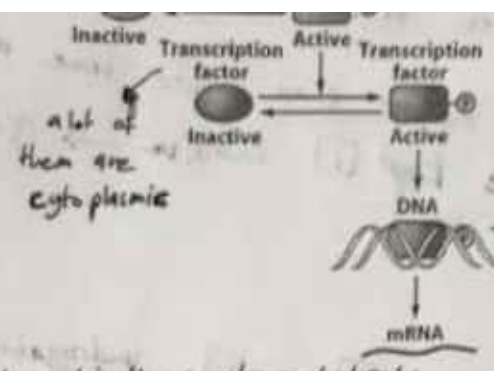


binds to the promoter of the target gene to activate transcription.

phosphorylation can activate a protein.
 the protein has a shape. Addition of phosphate group can induce conformational change.
 if it's an enzyme, the Addition will expose its active site → and then binding to the substrate

the used to be cytoplasmic, get phosphorylated, then this transcription factor got into the Nucleus and

of phosphate group
can induce conformational
change.
if it's an enzyme,
the Addition will
expose its active site



binds to the promoter of the target gene to activate transcription.

15.2 A Survey of Extracellular Messengers and Their Receptors (1)

- Extracellular messengers include:
 - Small molecules such as amino acids and their derivatives.
 - Gases such as NO and CO
 - Steroids
 - Eicosanoids, which are lipids derived from fatty acids.
 - Various peptides and proteins

the used to be cytoplasmic, get phosphorylated, then this transcription factor got into the Nucleus and activated transcription

to go to the Nucleus you have to have NLS.

NLS was masked when the transcription factor got phosphorylated it exposed the NLS then importin α and binds to it, take it inside.

is a result of 4 steps

imagine that the tumor cell can make this all the time phosphorylated. what happens? all the time transcription and activation of genes. (a lot of these Kinases mutated, which means active all the time)

* 50% of cancer drugs against Kinases.

in the cell there about 500 Kinases.

to activate them.

1 → 2 → 3 → 4 → 5 → 6

→ all these steps happen to 6 becomes active

If  6 is already on (active)

you don't need all the previous steps.

- target for modulation
 - target for tumor cells
 - target for therapy
-  Things are fighting over this target.
want to keep it active all the time
the drug wants to block the phosphorylation process.

* If you want to design a drug to target  the pathway to be activated. Cells tend to be efficient (they do the best job with the limited resources or activities). So rather than waste time and target 1, 2, 3, 4, 5 we target only . If we keep  inactive → Nothing will happen after.

Sometimes, you might not have access to  or it's undruggable. (Cisplatin)

A target signal binds to the Receptor, activates a pathway. In order to activate the pathway signal binds to the Receptor then there will be signal transduction, which means the transmission of that specific signal into the cell by some sort of conformational change within the receptor → activates a pathway → there will be multiple steps involving modulation, signal amplification → Action

5/1/2018

A Survey of Extracellular Messengers and Their Receptors (2)

- Receptor types include:

1) – G-protein coupled receptors (GPCRs)

2) – Receptor protein-tyrosine kinases (RTKs)

3) – Ligand gated channels or ion channels opens and closes based on the binding to molecules

← 4) – Steroid hormone receptors

5) – Specific receptors such as B- and T-cell receptors

↓ ↓
T-helper T-cytotoxic

within the
cell
(intracellular
receptors)

15.3 G Protein-Coupled Receptors and Their Second Messengers (1)

- G protein-coupled receptors (GPCRs) are numerous.

- GPCRs have seven transmembrane domains and interact with G proteins.

we have a receptor and G-proteins but we're talking about a class of G-proteins (α, β, γ) $\rightarrow$ large proteins

3 subunits $\rightarrow$ heterotrimeric

How do we know we have 7 transmembrane domains?? in a protein??

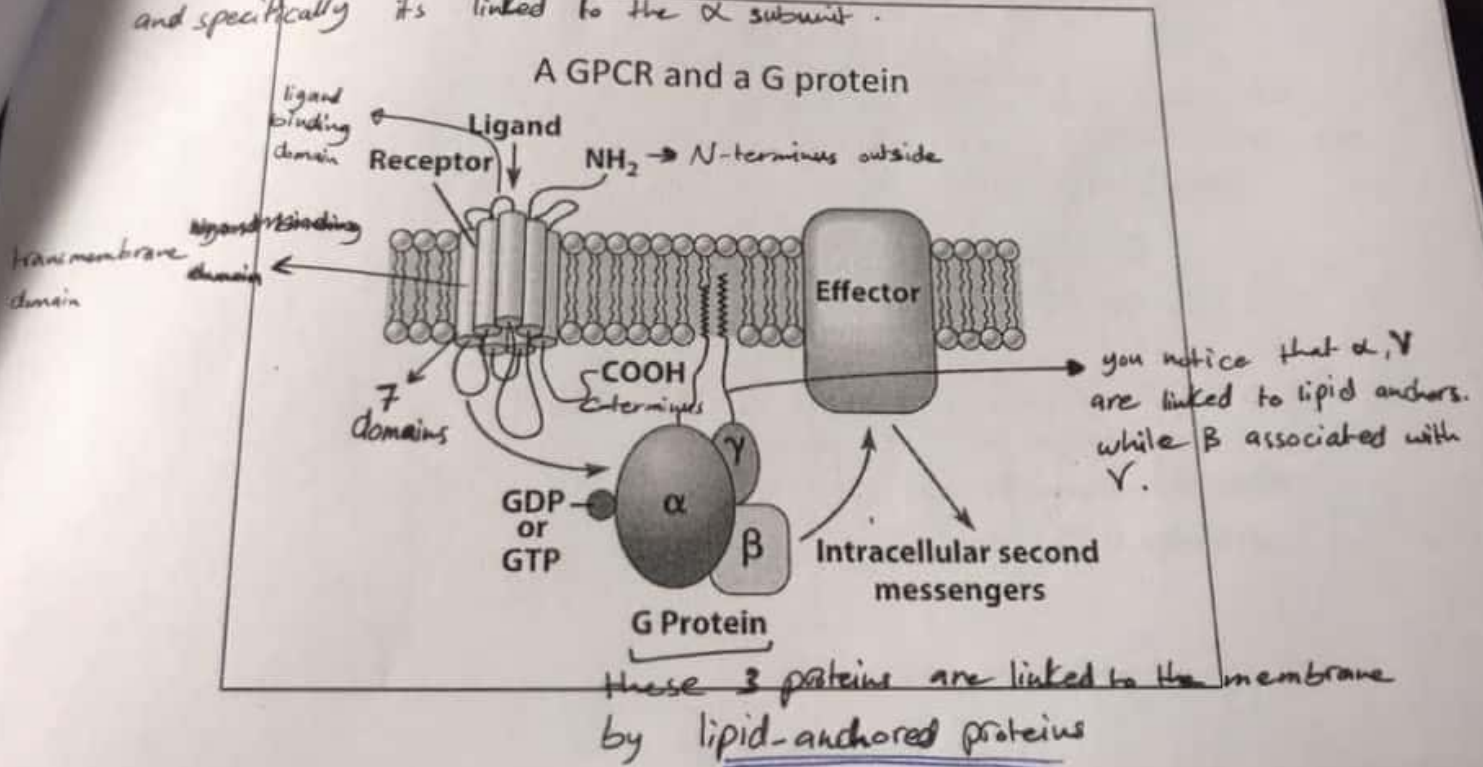
Hydropathy plot. amino acids
7 ΔG^+ $\rightarrow$ 7 hydrophobic stretch
20-25

prim ER 100% v2
is stop transfer q2 v2
7 stop transfers is v2 8

The drugs are targeting G-protein coupled receptors because they're involved in a number of diseases. So they take these molecules and design a drug so that you can use them for the treatment of different diseases.

5/1/2018

- we call these G proteins, because they are linked either to GDP or GTP and specifically it's linked to the α subunit.



- before the ligand is bound, the GPCR and the G proteins are separate from each other.

these steps:-

ligand bound to the receptor → the Receptor somehow has to collect the G-proteins → which later can activate the effector

- If for some reason, the effector is turned on, you don't need the previous steps (upstream).

Adrenaline:- Heart beat goes up

Heart Rate:- $\frac{\text{Number of beats}}{\text{per minute}}$

Cardiac output $\downarrow$ $\rightarrow$ $\frac{\text{Cardiac output}}{\text{Stroke volume}}$

5/1/2018

- Sometimes when you are in a situation "Fight or flight". They are related to epinephrine and Norepinephrine.

Examples of GPCRs and their ligands

you can increase the heart cardiac output, which means that the flow of the blood $\uparrow$ when it increases it goes to the tissues, and there $\rightarrow$ metabolically active $\rightarrow$ produce oxygen $\rightarrow$ for cells

TABLE 15.1 Examples of Physiologic Processes Mediated by GPCRs and Heterotrimeric G Proteins

Stimulus	Receptor	Effector	Physiologic response
Epinephrine	β -Adrenergic receptor	Adenyl cyclase	Glycogen breakdown
Serotonin	Serotonin receptor	Adenyl cyclase	Behavioral sensitization and learning in Aplysia
Light	Rhodopsin	cGMP phosphodiesterase	Visual excitation
IgE-antigen complexes	Mast cell IgE receptor	Phospholipase C	Secretion
EMT Peptide	Chemotactic receptor	Phospholipase C	Chemotaxis
Acetylcholine	Muscarinic receptor	Potassium channel	Slowing of pacemaker activity

Adapted from L. Strayer and H. R. Bozma, reproduced with permission from the Annual Review of Cell Biology, vol. 2, © 1986, by Annual Reviews Inc.

- Epinephrine, Norepinephrine: They function through 1 Receptor $\rightarrow$ β -Adrenergic receptor $\rightarrow$ G protein coupled Receptor

* Glycogen breakdown:- produces glucose $\rightarrow$ ATP synthesis

* Heart Rate $\uparrow$ why? $\rightarrow$ blood $\downarrow$ oxygen $\downarrow$

blood $\downarrow$ oxygen $\downarrow$

$\rightarrow$ Glucose $\rightarrow$ Cells $\downarrow$ $\rightarrow$ Cell Resp

Action $\leftarrow$

Action

- Epinephrine, Noradrenaline: They function through
- 1 Receptor $\rightarrow$ β -Adrenergic receptor $\rightarrow$ G protein coupled Receptor
- * Glycogen break down $\rightarrow$ produces glucose $\rightarrow$ ATP synthesis
- * Heart Rate $\uparrow$ why? ^{adrenaline} _{oxygen}

blood $\uparrow$
oxygen $\downarrow$

Cells $\uparrow$

Glucose

Cellular
Respiration

ATP

G Protein-Coupled Receptors and Their Second Messengers (2)

- Signal Transduction by G Protein-Coupled Receptors
 - Ligand binding on the extracellular domain changes the intracellular domain.
 - Affinity for G proteins increases, and the receptor binds a G protein intracellularly.
 - GDP is exchanged for GTP on the G protein, activating the G protein.
 - One ligand-bound receptor can activate many G proteins.

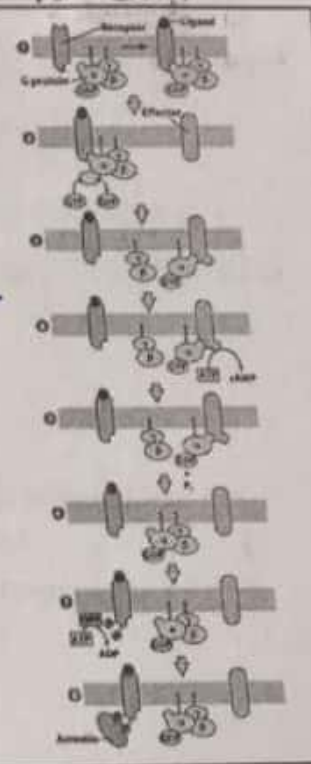
Regulation

functions
through
GPCR

- the pacemaker $\rightarrow$ Regulates the Number of beats per minute.
- the potassium channel (opening or closing) located within the membrane

1) ligand ~~is~~ ^{is not} bound
 2) ligand is bound, if you notice at the inactive stage, the α, β, γ GDP is linked to the α subunit. Once the ligand is bound there will be a switch from GDP $\rightarrow$ GTP associated with 5/1/2018
 So the ~~the~~ ligand bound to the Receptor $\rightarrow$ the Receptor will Recruit the G-proteins $\rightarrow$ dissociate GDP for GTP.

Mechanism of receptor-mediated activation/inhibition by G proteins



α also $\leftarrow$
 $\beta\gamma$

Also, α subunit
 Now is GTP-bound
 to the Receptor dissociates
 from the $\beta\gamma$
 $\downarrow$
 exist as a dimer

And then the α
 subunit will try to
 activate the effector

(like adenylyl cyclase)
 $\rightarrow$ an enzyme that converts ATP $\rightarrow$ cAMP

G-protein coupled receptors and their

And then the α subunit will try to activate the effector



(like  adenylyl cyclase)

→ an enzyme that converts $ATP \rightarrow cAMP$

G Protein-Coupled Receptors and Their Second Messengers (3)

- Termination of the Response
 - *Desensitization* – by blocking active receptors from turning on additional G proteins.
 - *G protein-coupled receptor kinase (GRK)* activates a GPCR via phosphorylation.
 - Proteins called *arrestins* compete with G proteins to bind GPCRs.
 - Termination of the response is accelerated by *regulators of G protein signaling (RGSs)*.

$cAMP \rightarrow$ it's a second messenger $\rightarrow$ Amplify the signal

* ~~is~~ ^{is} anytime you will activate a pathway, you have to find a way to shut it. ~~is~~ ^{is} the α subunit GTP bound all the time, because at a certain point you don't want to stay in the fight stage. you have to go back to the normal.

One thing that can happen that you want to try to get the GTP from α subunit and Revert it to GDP

- phosphodiesterase:- $\dagger$ breaks the cAMP $\rightarrow$ levels go down
or degrades
- arrestin:- Can bind to the α subunit and shut it down

Phosphatidylinositol: "Another second messenger"
↓
another component of the plasma membrane

It's a carbohydrate linked to a lipid within the plasma membrane 5/1/2018
How is phosphatidylinositol Can also act as a second messenger??
the second messengers in general, their function is Amplification.

G Protein-Coupled Receptors and Their Second Messengers (5)

- Phosphatidylinositol-Derived Second Messengers
 - Some phospholipids of cell membranes are converted into second messengers by activated phospholipases.
- Phosphatidylinositol Phosphorylation

4)

IP₃ Receptor or Ryanodine Receptor

↓
non
invariant
3 phosphates

So generation of the IP₃ from PI molecules

Molecular Basis:- what are the molecules involved in the pathway or action-

split of the PI → leads to the production of IP₃ + DAG

↓
One of the things they can do is to bind to the IP₃ Receptors on the smooth ER and open the calcium channels within the smooth ER → Release Ca²⁺

→ can act through another mechanism to activate different pathways.

One of the things they can do is to bind to the IP₃ Receptors on the smooth ER and open the calcium channels within the smooth ER → Release Ca^{+2}

activate pathways.

- activation of one enzyme :- phosphatidylinositol phospholipase C activates the kinases that split the PI at the plasma membrane.

- The phosphorylated phosphoinositides form lipid-binding domains called **PH domains**.

3)

in the plasma membrane

Calcium channels they open and close.

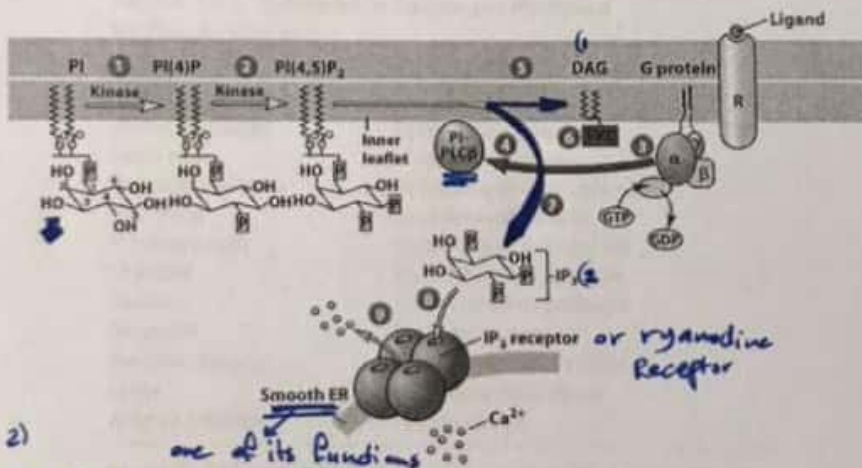
So when they open they allow Ca^{++} to get in from extracellular fluid into cells.

you use what you use. but you have extra. the extra you store them in the smooth ER. after you store them, at a time you need them.

1) through different kinases phosphorylates it. it can be split into 2 different molecules:-

DAG
IP3

The generation of second messengers as a result of breakdown of PI

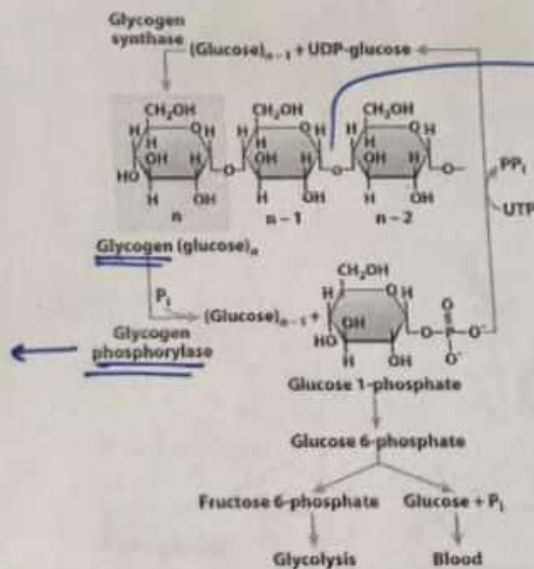


2) one of its functions is that they act as a storage compartment for the Calcium.

- the Ca^{++} is needed for the cell for many things;- muscles contraction, proteins —

Glycogen:-
muscles + liver

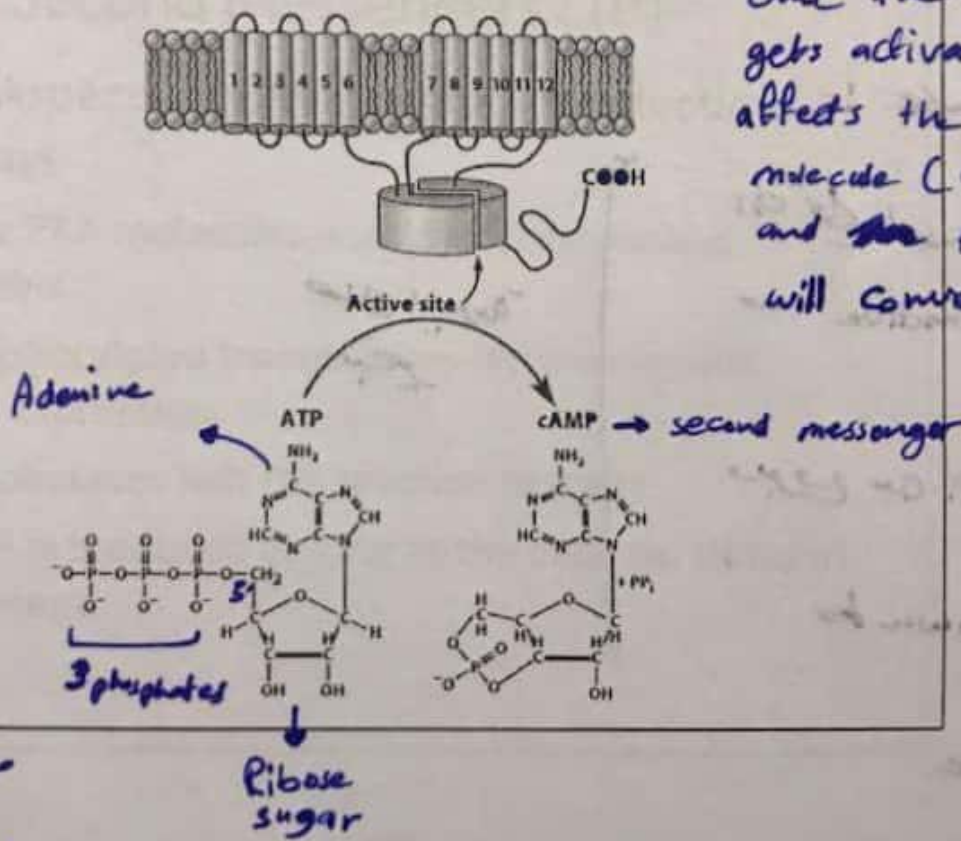
The reactions that lead to glucose storage or mobilization



Glycosidic bond:
the bond between glucose
molecules.

this enzyme activated
through a GPCR
pathway. Glycogen
phosphorylase somehow
tries to break
down the glycosidic linkage

Formation of cAMP from ATP



Once the GPCR gets activated it affects the effector molecule (adenylyl cyclase) and the effector will convert ATP → cAMP

Glucagon :- hormone

2a

Ribose sugar

The response by a liver cell to glucagon or epinephrine

Glucagon :- hormone
Glucose

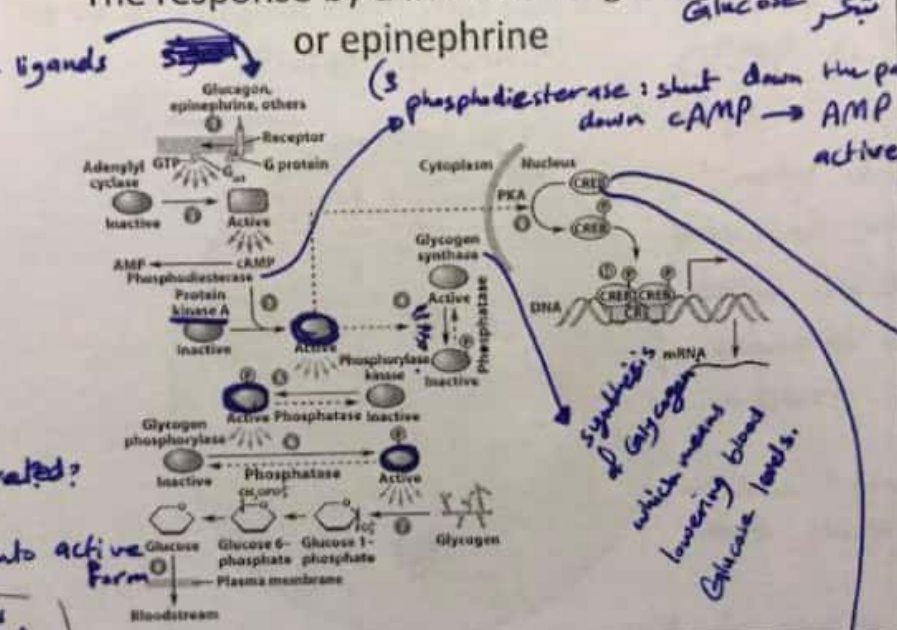
they work (2) through GPCR

stimulated which means GTP bound. Adenyl cyclase is inactive. How does it get activated?

G_q converts it into active form. Stimulus: that means the active form of it

Once Adenyl cyclase gets activated → ATP → cAMP

2a



(inhibits) Cytoplasm) by the reaction of PKA which translocated them to the nucleus they get phosphorylated.

phosphodiesterase: shut down the pathway by breaking down cAMP → AMP → cyclic active

Cyclic AMP Response element binding protein (transcription factors) binds to the DNA they are called (transcription factors) to activate the transcription of target genes

4) If you want to continue there's a molecule within ~~from~~ cells that ~~and~~ → protein Kinase A

↓
involves phosphorylation

How? they somehow
act on a molecule
like: phosphorylase

kinase → Normally it's inactive.

How do you activate it?

Any time you're talking about activation of an enzyme it's all about conformational change, so you expose its active site then the substrate binds.

So, Phosphorylase Kinase was inactive, gets activated by Kinase A.

↓
reaction of protein

5)

Signal response :-

~~metabolism~~

Short term :-

زیر لفظی حدستی حدی
نزل اسر عند حدی .



long term :-

Carbohydrate 1
Metabolism

سریع 1
بقرعہ طریقه 1
salivary
amylase

اللی حدی 1
Oral cavity 1

breaking
down

small
Intestines

8) in this case, transcription and translation of the enzymes required for regulation of Glucose and ~~Glycogen~~ Glycogen

this is the long term response

transcription + translation ← دیر قیامی

Short response :-

Glycogen phosphorylase دیر قیامی
active ← inactive

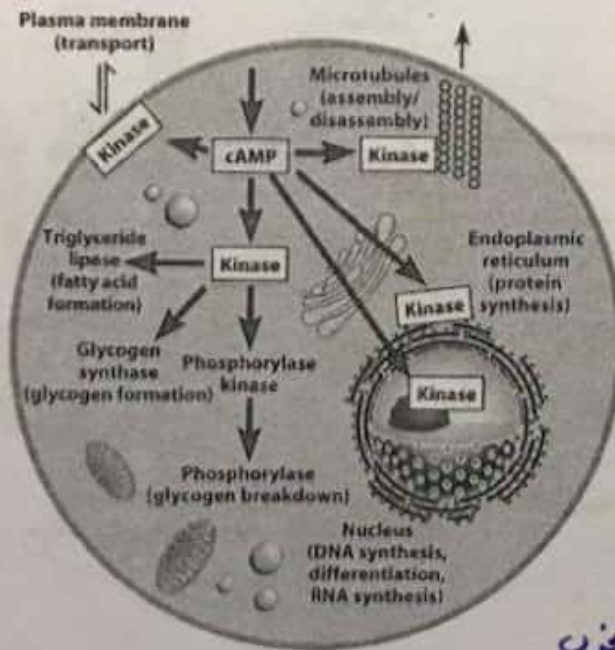
Glycogen دیر قیامی

↓
Break down to

↓
Glucose

صاف بصری
بصری

The variety of processes that can be affected by changes in [cAMP]



CAMP can cause so many things.

So many different things ~~can~~ that second messenger ~~does~~ can do when it's generated by one effector molecule.

- Once you get this, so many things get activated

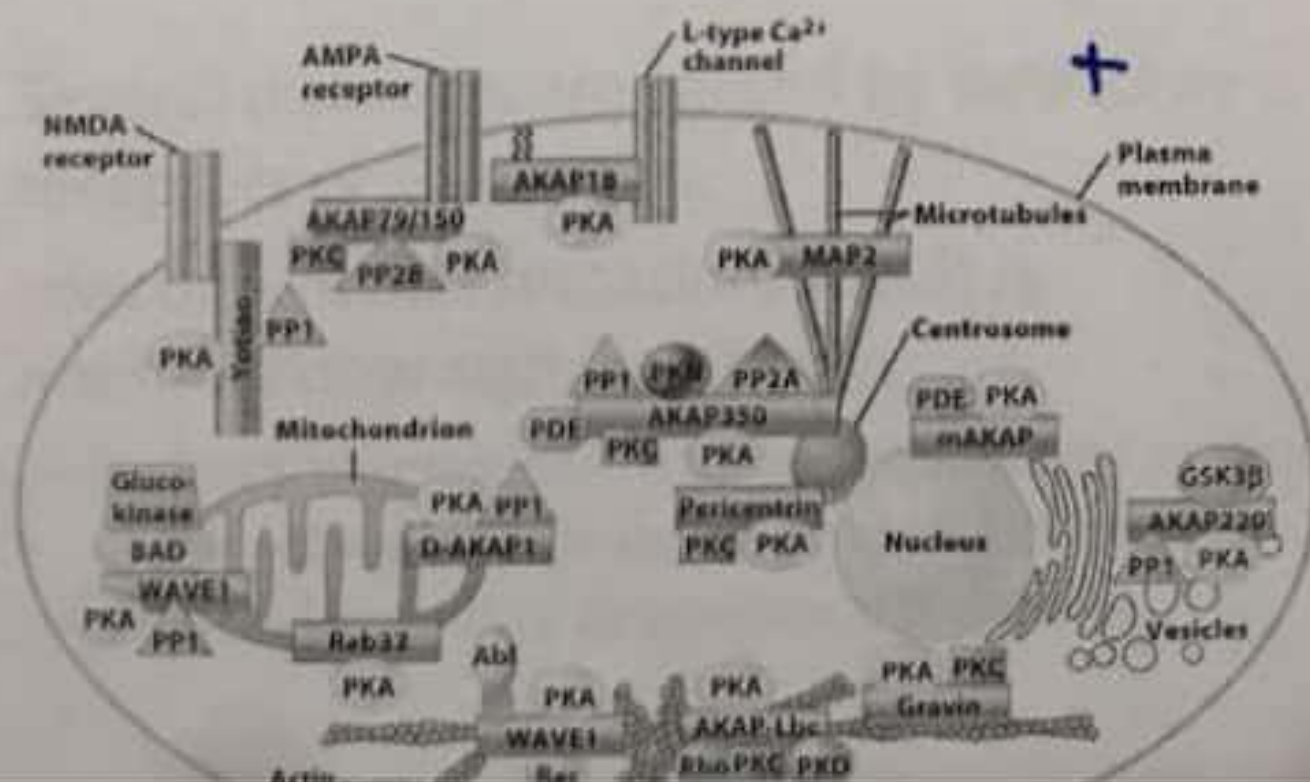
عندما تحصل على هذا اذا ضربت اشياء كثيره
كلها.

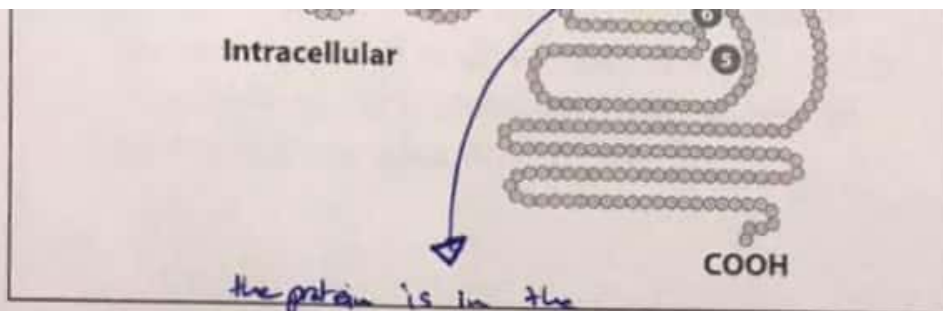
There's a complementation within the cell, if you can't do ^{it} ₁₉ one way, there are so many other ways that you can do it.

Tissue	Hormone	Response
Liver	Epinephrine and glucagon	Glycogen breakdown, glucose synthesis (gluconeogenesis), inhibition of glycogen synthesis
Skeletal muscle	Epinephrine	Glycogen breakdown, inhibition of glycogen synthesis
Cardiac muscle	Epinephrine	Increased contractility
Adipose	Epinephrine, ACTH, and glucagon	Triacylglycerol catabolism
Kidney	Vasopressin (ADH)	Increased permeability of epithelial cells to water
Thyroid	TSH	Secretion of thyroid hormones
Bone	Parathyroid hormone	Increased calcium resorption
Ovary	LH	Increased secretion of steroid hormones
Adrenal cortex	ACTH	Increased secretion of glucocorticoids

آزاد بیه

PKA-anchoring protein signaling





the protein is in the plasma membrane.

but the ~~trans~~ mutation is located within the transmembrane segment of the protein

Human diseases linked to the G protein pathway

TABLE 1 Human Diseases Linked to the G Protein Pathway

Disease	Defective G protein*
Albright's hereditary osteodystrophy and pseudohypoparathyroidism	G_{10}
McCune-Albright syndrome	G_{11}
Pituitary, thyroid tumors (gsp oncogene)	G_{10}
Adrenocortical, ovarian tumors (gip oncogene)	G_{12}
Combined precocious puberty and pseudohypoparathyroidism	G_{12}
Disease	Defective G protein-coupled receptor
Familial hypocalcaemic hypercalcaemia	Human analogue of BoPCAR1 receptor
Neonatal severe hyperparathyroidism	Human analogue of BoPCAR1 receptor (homozygous)
Hyperthyroidism (thyroid adenomas)	Thyrotropin receptor
Familial male precocious puberty	Luteinizing hormone receptor
X-linked nephrogenic diabetes insipidus	V2 vasopressin receptor
Retinitis pigmentosa	Rhodopsin receptor
Color blindness, spectral sensitivity variations	Cone opsin receptor
Familial glucocorticoid deficiency and isolated glucocorticoid deficiency	Adrenocorticotrophic hormone (ACTH) receptor

*As described in the text, a G protein with a G_{10} acts to stimulate the effector, whereas a G protein with a G_{11} inhibits the effector.
Source: G. E. Clapham, reproduced with permission from Nature, vol. 377, p. 198, 1994. © Copyright 1994, by Macmillan Magazines Ltd.

mutation of the hydrophobic α is amino acids $\rightarrow$ hydrophilic which disrupts the transmembrane domain stretch as 3, 4, 5 of hydrophobic amino acids

bio
pro
r

15.4 Protein-Tyrosine Phosphorylation as a Mechanism for Signal Transduction (1)

- **Protein-tyrosine kinases** phosphorylate tyrosine residues on target proteins.
- Protein-tyrosine kinases regulate cell growth, division, differentiation, survival, and migration.
- **Receptor protein-tyrosine kinases (RTKs)** are cell surface receptors of the protein-tyrosine kinase family. *part of the plasma membrane*

there is other class of Receptors:-
* **Non-Receptor tyrosine Kinases:**
they are not part of the plasma membrane
they're intracellular ~~kinases~~ tyrosine Kinases

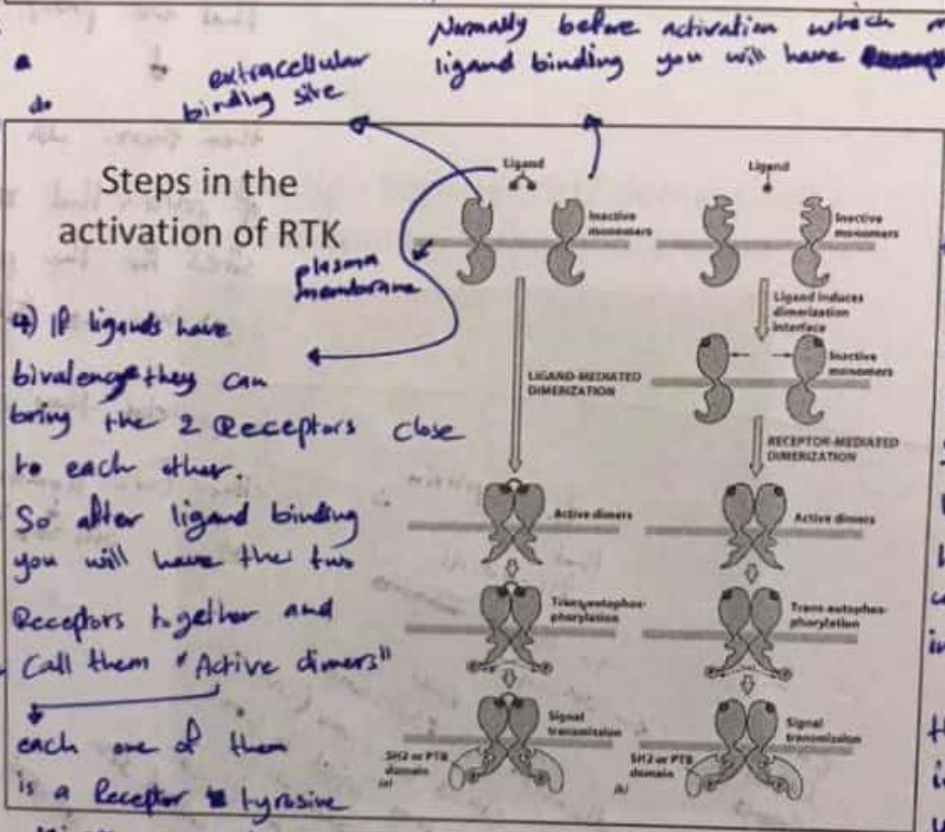
the amino acid that's involved in this process because we're talking about phosphorylation and the tyrosine has OH group so it can phosphate can be attached

to ~~OH~~ hydrogen bond in the OH group.

3 amino acids that can be phosphorylated:- serine, tyrosine, Threonine.

2) every cell has plenty of Receptors & a cell response to a signal because they have Receptors

1) Normally before activation which means that before ligand binding you will have ~~inactive~~ Receptors



4) If ligands have bivalence they can bring the 2 Receptors close to each other. So after ligand binding you will have the two Receptors together and we call them "Active dimers" each one of them is a Receptor & tyrosine Kinase. So what they do?? they phosphorylate targeting proteins.

3) - as you can see the ligand is involved it has bivalent binding → (2 binding sites)

So, normally before ligand binding you have Receptors, as what we call → inactive monomers

they are present in single. unlike GPCR

they pass the membrane once and they have 2 binding sites for multiple ligands

when they are "inactive monomers" they have 5)
Basal level of phosphorylation $\rightarrow$ they can phosphorylate
themselves. so if you phosphorylate your self $\rightarrow$ Auto phosphorylation

but ~~when~~ Normally when they are present in single Receptor they
have a very low level of phosphorylation that doesn't activate the
pathways.

bivalence ligands bring the RTKs together $\rightarrow$ trans-~~auto~~ phosphorylation

$\downarrow$
they phosphorylate
each other

$\downarrow$
increase the phosphorylation

$\downarrow$
the tyrosine gets phosphorylated

so will be an increase in
the Number of tyrosines
that are phosphorylated.

$\downarrow$
then there will be recruitment

that are phosphorylated.



then there will be recruitment of protein that has binding sites for the phosphorylated tyrosine → (SH2 or PTB) domains



phosphorylated
tyrosine binding
domains

proteins they do have

either (src Homology domain)

or

SH1 SH2 SH3

first protein
that was
confirmed as
an ~~oncogene~~

← oncogene

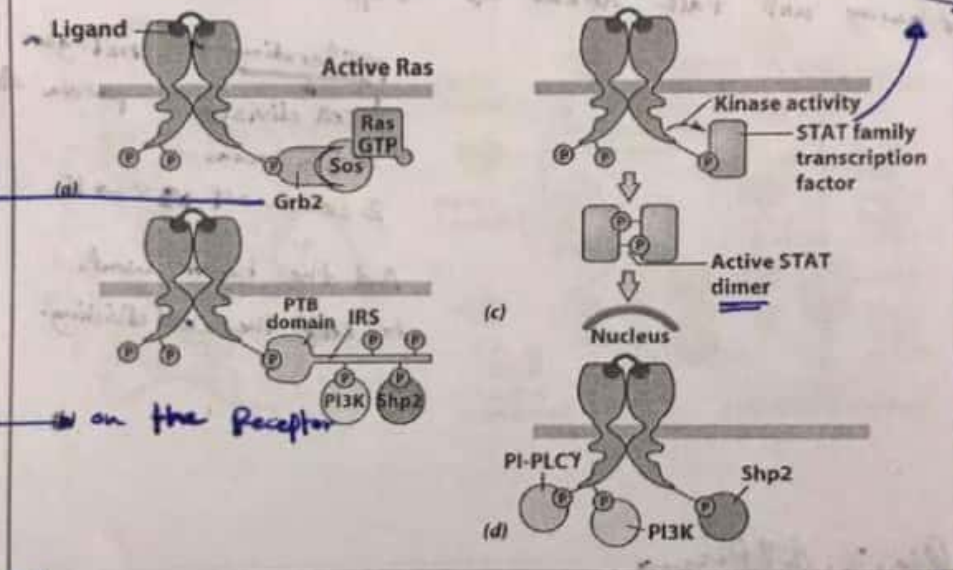
Cancer
Causing
Genes

It was discovered in sarcoma
viruses which causes sarcoma (cancer)
in chickens. when they tried to locate
the gene that's responsible for
causing the transformation of cells
it turned out to be a virus called src

Later they found a domain in several other
proteins similar to the src.

when there's an increase in phosphorylated tyrosines
you recruit proteins that have either SH2 ~~to~~ binding domain
or PTB ~~to~~ domain and they to ~~these~~ phosphorylated tyrosines.

A diversity of signaling proteins



STAT: (signal transducer and activator of transcription)

they normally bind to the phosphorylated tyrosine. and what happens? they get dimerized. when they get dimerized they can be translocated into the Nucleus and activate the transcription by binding to certain transcription factors.

SH2 domain
PTB domain

Grb2 binds to the phosphorylated tyrosine. and at the same time binds the SOS protein.

there's a protein called

Ras → It's a small G protein, Ras is Guanosine associated either GTP or GDP

Normally without ligand binding and Receptor activation

Ras GDP state

Protein-Tyrosine Phosphorylation as a Mechanism for Signal Transduction (4)

- The Ras-MAP Kinase Pathway

binds the SOS protein.

there's a protein called

Ras → It's a small G protein, Ras is Guanosine associated either GTP or GDP

Normally without ligand binding and Receptor activation.

Ras GDP state

which makes inactive.

Ras it has to be activated by switching GDP → GTP

How do you know this?

Sos: Son of sevenless was a gene discovered in drosophila. It happens that this gene has a homolog in humans and mammals.

Ras: switch the GDP to GTP. So it called Guanine Nucleotide exchange factor. GEF.

SOS binds to Grb2 and at the same time binds to Ras



binding to certain transcription factors.

Protein-Tyrosine Phosphorylation as a Mechanism for Signal Transduction (4)

• The Ras-MAP Kinase Pathway

- Ras is a G protein embedded in the membrane by a lipid group.
- Ras is active when bound to GTP and inactive when bound to GDP.

If have Ras-GTP state all the time. what does that mean?
we don't need ⁱⁿ all the previous steps.

1. Ras 1 mutation (loss of GTPase)

39-40% of cancers have mutations in Ras

It's always active $\rightarrow$ in the GTP state

If it stays active all the time this is bad, because if we continue in the pathway will talk about $\rightarrow$ "steps of generalized MAP kinase cascade"

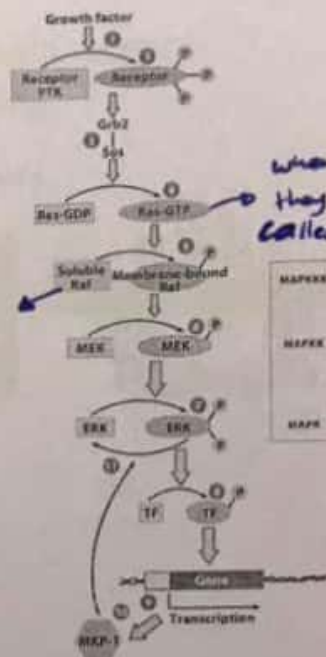
proliferation $\rightarrow$ cell division
which means

2 cells $\rightarrow$ 4 $\rightarrow$ 8 $\rightarrow$ 16

And the tumor wants to keep the cell dividing.

The steps of a generalized MAP kinase cascade

proliferation $\rightarrow$ $G_1 \rightarrow S \rightarrow G_2 \rightarrow M$
which means cell division



Mitogen activated protein Kinase Kinase Kinase

when it becomes active they go after a molecule called Raf = equivalent for MAPKKK

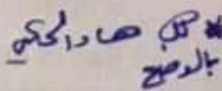
and Raf becomes active by being phosphorylated.

because it's a Kinase it can phosphorylate a downstream Kinase $\rightarrow$ MEK is equivalent for MAPKK.

And also MEK₂₈ can phosphorylate another ~~kinase~~ extracellular Regulating Kinase

$\rightarrow$ called ERK or MAPK becomes phosphorylated $\rightarrow$ active then the Transcription Factor (TF) will get phosphorylated $\rightarrow$ active Once they are active they bind to the target gene through the promoter $\rightarrow$ activate the transcription of gene required for cell cycle ($G_1 \rightarrow S \rightarrow G_2 \rightarrow M$) which means cell division.

*² ;
imagine that ~~if~~ if it's a tumor cell that
has a mutation in Ras $\rightarrow$ locked in the GTP state $\rightarrow$ which means
active all the time $\rightarrow$ production of genes required for cell cycle
progression $\rightarrow$ which means that the cell keeps dividing without control.

requires ligand $\leftarrow$ 
binding

or the receptors
are already close
to each other.

if you want to prevent this from the beginning. what will you do?
block the access to the tyrosines that get phosphorylated

So a lot of the times they switch the Tyrosine with Alanine $\rightarrow$ because
they don't have OH so they don't get phosphorylated. if this doesn't happen
you stop down the pathway.

Protein-Tyrosine Phosphorylation as a Mechanism for Signal Transduction (5)

- Ras-MAP kinase pathway (continued)

← Accessory proteins play a role:

- 1 • GTPase-activating proteins (GAPs) shorten the active time of Ras.
- 2 • Guanine nucleotide-exchange factors (GEFs) stimulate the exchange of GDP for GTP.
- 3 • Guanine nucleotide-dissociation inhibitors (GDIs) inhibit release of GDP.

switch the GDP
to GTP

Receptor Tyrosine
Kinases

q. 10 to 12

Class of Receptors

q. 10

Called serine-Threonine Kinases
Receptor

It means that Not

q. 11

tyrosine ~~that~~ that gets phosphorylated, but serine and Threonine

~~Among them~~

we call them JAK - STAT Receptors
↓
Janus activated Kinases

Functions - Can bind the ~~stat~~ STATs, then go and bind to transcription
Factors --- and so on.
~~also called~~

of glucose.

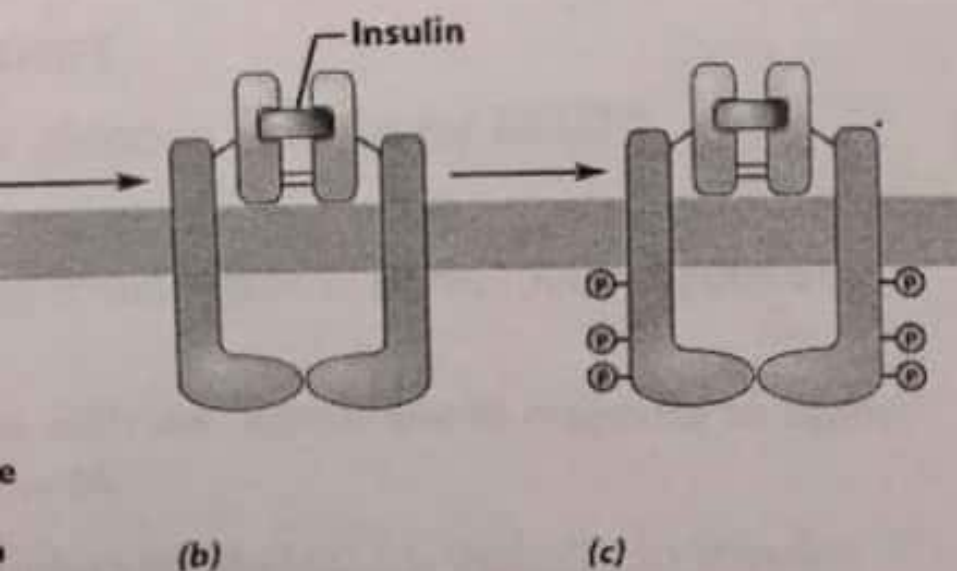
Receptor is a protein-tyrosine kinase

Phosphorylated receptor associates with insulin → substrate proteins (IRSs).

Proteins with SH2 domains, which activate signal molecules.

Containing proteins are kinases that use a lipid, PI 3-kinase (PI3K).

Response of the insulin receptor to ligand binding

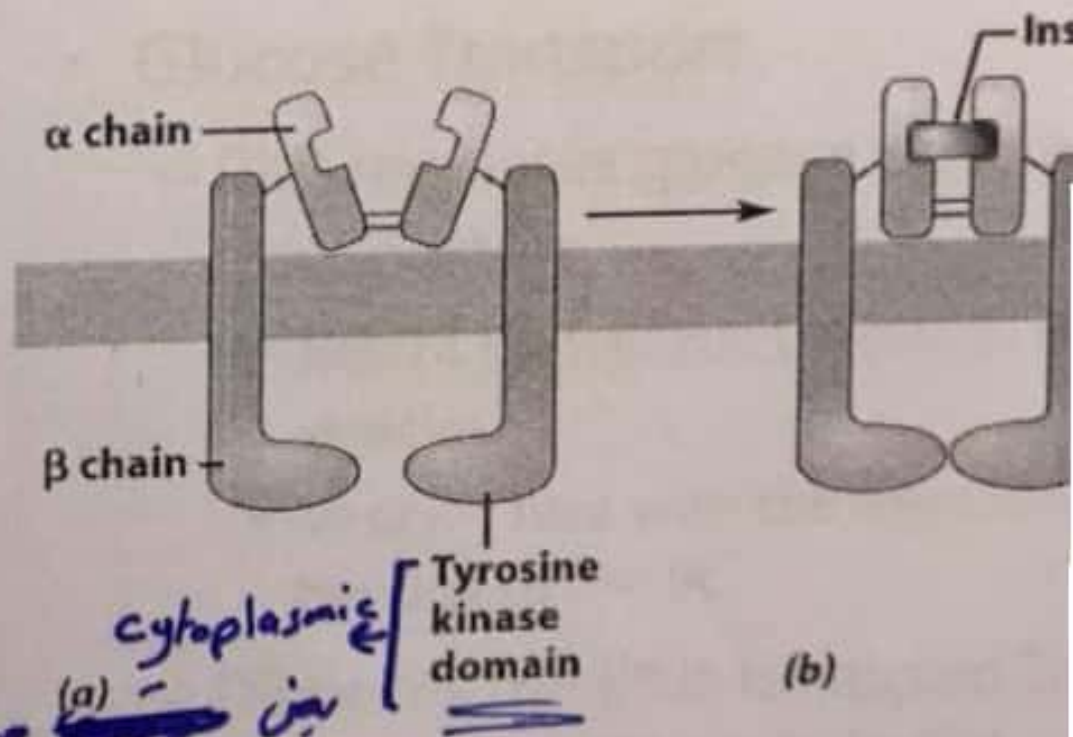


is an example molecule that has a tyrosine kinase domain and its Cytoplasmic.

what does that mean?
insulin Receptor substrate (IRS) when they bind they dimerize,

increase the phosphorylated tyrosine and activate a pathway similar to the MAP Kinase pathway and also similar to the Receptor tyrosine kinase pathway

The response of the ins to ligand bind



cytoplasmic
domain / (a) بين دأط الخلية
(b)

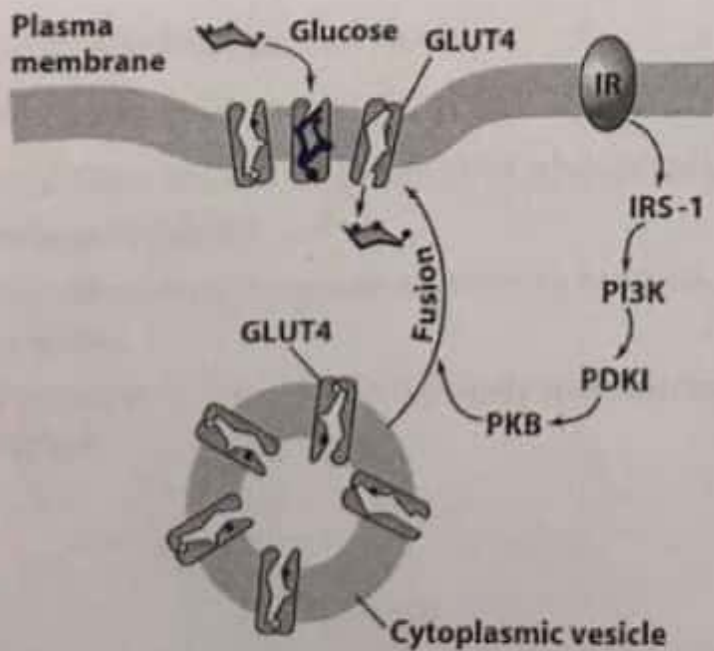
GLUT 1, 2, 3, 4

↓
Glucose transporter

diabetes (علاج) all mutations) no 2003

↳ A mutation in GLUT 1 or 2 or 3, 4
that are involved in Glucose 5/1/2018
transporting

Regulation of glucose uptake in muscle and fat cells by insulin



2 types of

Diabetes:-

1)

2) Adults form
of diabetes

There could be not
just insulin
problem. There
could be a problem
in the insulin
Receptor itself,
or could be a
problem in Glucose
transporters.

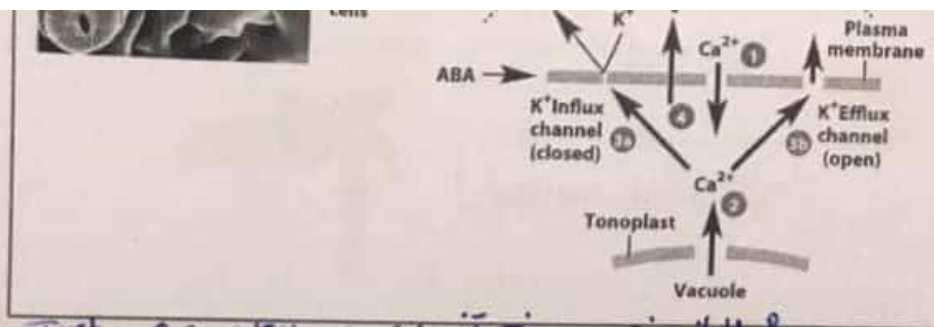
Protein-Tyrosine Phosphorylation as a Mechanism for Signal Transduction (10)

- Signaling Pathways in Plants

- Plants lack cyclic nucleotides and RTKs.
- Plants have protein kinases that phosphorylate histidine residues.
 - The downstream cascade is similar to MAP kinase cascade.
 - The target of the cascade is usually transcription factors.

15.5 The Role of Calcium as an Intracellular Messenger (1)

- Cytoplasmic calcium levels are determined by events within a membrane.
 - Calcium levels are low in the cytosol because it is pumped out into the extracellular space and the membrane is highly impermeable to the ion.
 - Calcium channels can be transiently opened by action potential or calcium itself.
 - Calcium binds to **calcium-binding proteins** (such as **calmodulin**), which affects other proteins.



Things are not just one way → closed → circuit →

Things are not interconnected

15.6 Convergence, Divergence and Crosstalk Among Different Signaling Pathways (1)

Signaling pathways can converge, diverge, and crosstalk as follows:

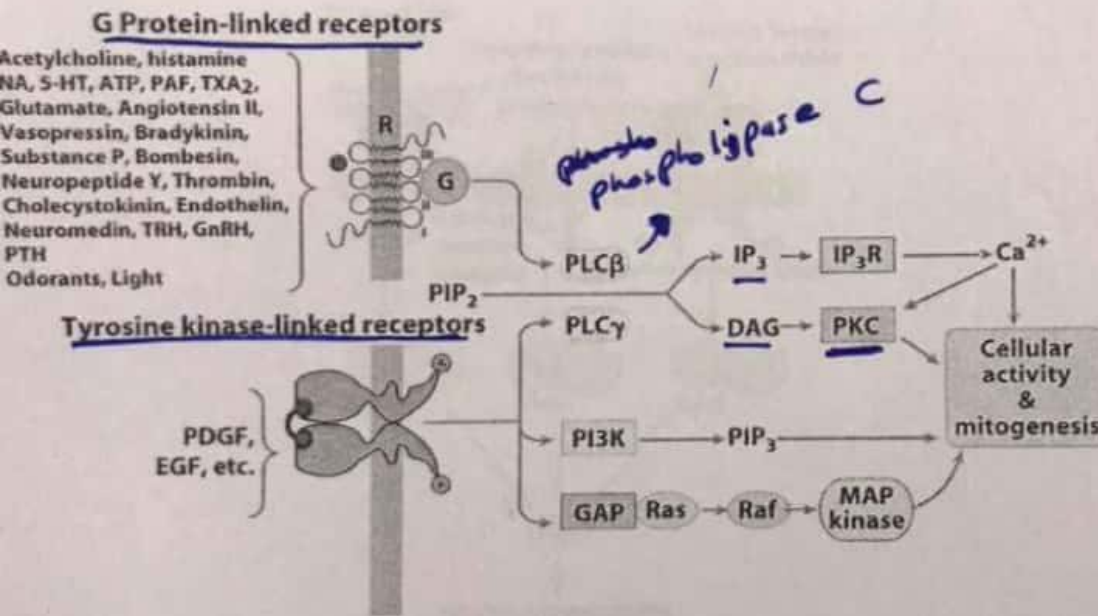
- Signals from unrelated receptors can *converge* to activate a common effector.
- Identical signals can *diverge* to activate a variety of effectors.
- Signals can be passed back and forth between pathways as a result of *crosstalk*.

There's an interaction between different pathways

different molecules converge on one target

One Receptor can activate multiple pathways

Examples of convergence, divergence, and crosstalk among signal transduction pathways



*Convert or proliferation
↓
proliferation*

يعمل
works through a receptor.
like steroids, because they
are hydrophobic they can't
cross the plasma
membrane, so they bind to their intracellular

5/1/2018

doesn't function
a receptor
because it's a
small gas. can
pass through
cells.

(1)
through

15.7 The Role of NO as an Intracellular Pathway

- Nitric oxide (NO) is both an extracellular and intercellular messenger with a variety of functions.
- NO is produced by nitric oxide synthase.
 - NO stimulates guanylyl cyclase, making cGMP.
 - cGMP decreases cytosolic calcium and relaxes smooth muscle.

(2) – NO also plays a role in male arousal.
Function ↓ drug Viagra: for the treat of
through

receptors
and they function
as transcription
factor that can
activate target
genes

Glycerin:-
has active
material in it

Through

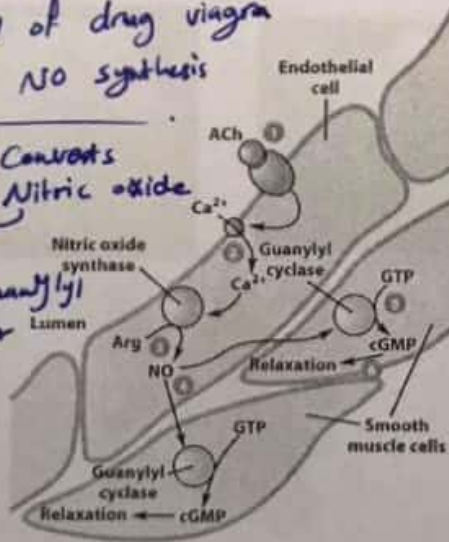
Glycerin -
the active
material in it
is NO

Signal transduction by means of NO and cGMP

③ the discovery of drug viagra
was basically NO synthesis

NO synthase converts
Arginine into Nitric oxide

Functions on Guanylyl
cyclase (similar
to Adenylyl
cyclase)
GTP →



↓ 5) ●
Causes smooth
muscles to

you 4) want to try to relax your blood vessels → there will be
more blood flow.

Relax a little
bit, which increases the diameter

of blood vessels → more blood flow

* the company that makes viagra drugs?
Pfizer

* [The cell biology of cancer] when cells go bad:

slide 2

* cells so bad, when they divide without control.

cancer / tumors → Both are bad "cancer is more"

- Something that is in the beginning of process, sometimes it's go to certain place.

→ cancer is has the potential "may not go" to go to different places.

↳ it's cells dividing without control. "tumors"

think about cell cycle [interphase - M]

→ repair things → G₁, S, G₂ → check and then mitosis (20-24hr) + (2hr)
↳ replication for DNA

* Tumors: uncontrolled proliferation of cells, something happen to cell that make them divide without control. * [sequences of genes] → mutation

① tumor suppressor genes ② onco genes ③ DNA repair genes

→ Benign cancer → stay confined

→ Malignant cancer → has the potential to go to different places.

- cancer incidence &

slide 3

↳ How many case of cancer are diagnose.

* [each 5 years]

Male → prostate cancer

Female → mostly breast cancer, cervical cancer

→ How many case of cancer are diagnose.

→ [each 5 years]

Male → Prostate cancer

Female → mostly breast cancer, cervical cancer

→ 99.9%, diagnosed based on if have Proinfection with HPV or not, (human papillomavirus)

* People realize that the only type of cancer that can be prevented [vaccination]

- (Early protection better management)

• ← receptor tyrosine kinases

bc ← then change in gene expression

* (tumor suppressor gene)

Stops

P53 ← guardian of genome

"stop mutations" transcription factor that activate certain molecule involve in cell cycle "stop" arrest to apoptosis.

* when problem happen in the cell

P53 gets stabilized (prevents from being degraded) then it's can act as transcription factor for stop the cell cycle.

and gets degraded very quick by [ubiquitination], then proteasome degrades.

* what happen in tumor cells, try to mutate P53 to prevents it work

ras → small (g-protein), [most "3-4%" of human tumor or cancer] have mutation on it.

(slide 4)

Just shows us, the cell interconnected [cross-dots] "things get interconnected in many pathways." * So all of these dots & are target for tumor cell, so that they can initiate or activate → leads to uncontrolled division [proliferation].

So the cell as "tumor" → will keep dividing.
Normally cells → stop dividing. [limited number of cell division].

* Tumor "hall marks" have limitless replicated potential. → can divided indefinitely.

(slide 5)

* Tumor → mass of cells. "angiogenesis" → creation of blood vessel.

→ our cells need limited number of blood supply, when become "trillions", so need more blood vessel. So tumor cells secrete [vascular endothelial growth factor] to create more blood vessels "need many oxygen-nutrient to supply" so more proliferation.

Tumor go in stages, stage 1 "can be remove", between stage 1 - II "Stage" (11 10 years)

→ progression for tumor

(slide 6) → normal cells & they have certain restrictions [anchorage dependent] need surface to stand.
all cells need contact to grow.

* tumor cells [anchorage independent], can grow in blood with need to attach.
contact inhibition "density inhibition".

- need surface to stand.
- * tumor cells [anchorage independent], can grow in blood with need to attach.
 - ① normal cells should have → contact inhibition "density inhibition".
means that when they contact to each other → stop dividing.
 - ② tumor cells lack → contact inhibition (prevent signalling between them that make them stop dividing)
 - * Normal "layers of skin" cells in tissue has an order "similarities in size and shape"
 - * cancer "tumor" in each cell cause variation.
 - ↳ Hyperplasia → growing cells without control "normal shape".
 - mild dysplasia → change shape, differentiation
 - carcinoma in situ (severe dysplasia) → cells becoming disorder, and shape is keep changing → still proliferate to their originates. ← benign.
 - cancer → cells start escape from where they originate, start to go somewhere else. ← malignant.
 - * How to classify tumor in tissue a carcinoma - sarcoma - leukemia, more frequently [cover everything in back] ← epithelial cells. Bone blood
 - (type) of cell a Adenomatous, squamous, myeloid, lymphoid
 - * (Define on cancer terminology + oncology) find tumor.
 1. tissue
 2. cells
 3. organs.

Slide 14

* Breast → * Lung → * Bone → Ewing's sarcoma & translocation between two chromosomes.

* Eye → retinoblastoma *

Slide 15 → diff between benign / malignant.

* they are confined

* very slow

* not considered cancerous.

↳ (abnormal cell growth not cancer) note.

↳ * start as benign then become invaded then go to different places.

* grow faster (have more mutation)

* metastasise (go to other tissues)

*(if not detected go on & become tumor)

Slide 17-18

Slide 19 → if normal mole get invaded → become "melanoma"

Slide 20 → process of metastatic & million cells have limited supply of blood vessels. what happens → these tumor cells start secreting [vascular endothelial growth factor] give more blood vessels by process (angiogenesis), (wide and leaky)

Slide 21 → the cell can escape from that and go to other places.

* There are things that are common to tumor cells & (from slide)

- All marks of cancer &

↳ feature and characteristic of tumor cells

"To become tumor"

↳ ① self sufficiency in growth factor. "signals"

↳ if no ① ⇒ problem in cell then cell death.

* signal "growth factor"

* paracrine signaling & cells affects nearby "adjacent" cells. ↳ make it bind outside the cell centre and secreted.

Slide 21-22

↳ Feature and characteristic of tumor cells "To become tumor"

① self sufficiency in growth factor. "signals"

↳ if no ① ⇒ problem in cell then cell death.

* Signal "growth factor"

* paracrine signaling & cells affects nearby "adjacent" cells. ↳ make & bind outside the cell centre and secreted.

② Insensitivity to growth-inhibitory signals

③ Evasion of apoptosis

* Tumor cells shut down the pathway

↳ programmed cell death that relate to apoptosis.

④ Limitless replicative potential.

↳ By activate the enzyme telomerase

"to keep dividing"

* Telomerase & nucleotides repeat multiple times at the ends of chromosomes

5 end problem ⇒ (replication problem), in every time cell divides it lose part of it.

because DNA polymerase require 3' OH ↳ put on it cap "telomerase" to protect them after (50-70 divides) lose the DNA ⇒ lose coding gene, here problem start.

So, the tumor cell & they activate telomerase enzyme, start working complementarily to telomerase and DNA polymerase.

So ⇒ Limitless replication potential.

always divid ⇒ cell times ⇒ make [transient cells with telomerase]

immortalization is

↳ repair "replace" damaged telomeres.

⑤ sustained angiogenesis.

↳ continuous every time tumor grows, [it give new blood vessel]

⑥ ability to invade and metastasize.

↳ go to different places and start again.

process involve migration

↳ to receive more oxygen/nutrient.

rec(5), (last)

process involve migration

rec(s), (last)

* cells go to choices 2

① divide - survive ② apoptosis "death signal"

- Evasion to apoptosis: should be death signal activation to signal to make cell die, tumor cell have mutation on p53 (no longer able to function), so don't go through apoptosis

* Matrix metalloproteinases & [MMPs] Extra cellular matrix, outside plasma membrane cells, one thing if activate the metastasize → there will be expression of a class of enzymes.

But (MMPs), they degrade the extracellular matrix, which allows cell to invade other tissues.

* one way to treat tumor is to → design inhibitor for (MMPs), (upregulation)? increase of expression, when they upregulated, so they have still the capacity to invade tissue.

So, some drugs are targeting (MMPs), and inactivate them and stop the pieces of metastasis.

(slide 20) → normal cell $\xrightarrow{\text{process}}$ tumor cell cancer is disease of aging. ^{for old people}
the more you live, the more that will exposed to environmental ^{why}
factors that will leads to cancer. #add/du

There is specific molecules called caspases "enzymes"

Slide 2.8 + 19

- normally they shouldn't be activated

* Initiator Pathway: there is signals, different molecules induced the cell to die because of death signal. There is ^{something} called death receptors

(Fas L / CD95L) — Ligands

Slide 20

↳ responding receptors "death receptors" — they have FADD • Fas-associated death domain.
(initiator caspase)

these have something "caspase 8" normally inactive

⊕ caspases activate each others "upstream caspase activated downstream caspases"
but caspase 8, have procaspase 8 "very limited caspases activity"
what's happen?

→ There will be correlation between (caspases 8) → then there will be instead of hundred caspase will be thousands, then they act to become (caspase 8+10).
then when they become active, they activation metabolic caspases [effector] 3A..

→ Effector caspase 8 they act on certain substrate → one of them lamin [A] Stabilize the nucleus.

→ So we need to de stabilize the nucleus "work on lamin A"

→ they destabilize the nucleus structure by degrading lamin A → nucleus fragmented then cell start die

→ There is two pathways of apoptosis: { extrinsic → secrete out side.
intrinsic → damage inside & radiation, ROS

→ effector caspase & they act on certain substrate → one of them lamin [A] Stabilize the nucleus

→ So we need to de stabilize the nucleus "work on lamin"

→ they destabilize the nucleus structure by degrading lamin A → nucleus fragmented then cell start die

→ There is two pathways of apoptosis: { extrinsic → secrete out side.
intrinsic → damage inside & radiation, ROS inside.

→ intrinsic & within mitochondria → change in membrane permeability, there is molecules inside the (intermembrane space) → will be go to outside (broke) then activate caspase, then to other caspase then go to death substrate finally apoptosis. [TTA FGG]

* cell divide, each time when divide lose some of telomerase, on 5' end. DNA Polymerase can't add nucleotides on 5', so don't know how to elongate it.

* Telomerase → Protein "enzyme" has complement to telomere. (slide 30)

In germ cells [telomerase] ← inactive, later tumor cell "limitless replicative potential"

→ (each time lose from telomere, will reduce) → so [telomerase] become active

(slide 31) * cell line → divide indefinitely. * primary cells → limited # of cell division.

- p53 → guardian of genome, mutated 50% of tumors, when p53 is mutated, the cells lose their ability to do apoptosis. → evade apoptosis.

* P53 → Transcription factor. (short-half-life) can't see in the cell.

* When become tumor & P53, act as to prevent tumor from function → So it's set stabilized by [phosphorylation] → they activate the transcription of P21

* when phosphorylated → stable "degradation"

CDK inhibitor

* CDK inhibitor's 8

↳ a cyclin-kinase inhibitor

They stop the cell cycle, go with progress (cyclin + CDK)

So P53 → activate P21 → then stop cell cycle

↳ induce apoptosis for cell cycle arrest. So tumor try to mutated it.

* Three classes of gene that target for tumor cells

(slide 35)

① tumor suppressor genes ② oncogenes ③ DNA repair genes

↳ stop [P53]

↳ go (BRCA1/2)

* Most of the cancers happen in somatic cell. "non-germline"

BRCA1 + BRCA2 → associated with 10-15% cases in breast cancer "hereditary"

↳ we can do mutation screen. if we have them, so we have the potential to have cancer. [sequence genome].

* The older you get, the more potential to have cancer's.

(slide 36)

* mutagenic → cause mutation. Become carcinogenic.

- HPV → cervical cancer,

(slide 37) E

* cancer tends to involve multiple mutations.

↳ colorectal cancer, as model of multiple mutation.

(slide 39)

* we can trace the tumor to its origin, and also follow the progression of tumor

* mutagenic → cause mutation. Become carcinogenic.

- HPN → cervical cancer,

(Slide 3) E

* cancer tends to involve multiple mutations.

↳ Colorectal cancer, as model of multiple mutation.

(Slide 59)

* we can trace the tumor to its origin, and also follow the progression of tumor in phenotype/genotype.

APE

(Slide 52) → to become [Hyperproliferative epithelium], there is chromosome on 5q region called (APC gene & adenomatous polyposis coli gene). this gene has mutation. from normal to From genotype → Phenotype, when the DNA become hypomethylated "methyl" group less on (CPG island) [Cytosine, Phosphate, Guanine] promoter regions.

So when DNA hypermethylated → become early adenoma. when Ras is mutated → become intermediate adenoma, then other gene (p53) get mutated so → become late adenoma, then mutated of p53 to → become tumor (Carcinoma)

[Pathology + gene] multiple gene, to know the progression of each step and gene's mutated to make diagnosis.

(Slide 61)

* How original can go to cancer gene?

① Mutation/deletion

② amplification

③ Chromosomal rearrangement

Philadelphia chrom.
Berkitt lymphoma

+ chromosomal rearrangement → Signal go to other place, become under the influence of other promoter → gene has expression different than normal situation.

receptor tyrosine kinases → if phosphorylated certain number become → proto-oncogene, so activation more and more for receptors. (slide 5)

+ overexpression → cancer = (dimerization) ← receptor binding with others.

④ slide 6 → 2/5

- if age 25 → refer to do mammogram

- cervical cancer → Pap smear (test) → cervical discharge, HPV. 2/5

- indication of having colorectal cancer → Fecal blood test

⑤ PSA "Prostate-specific antigen" → Prostate cancer, indicators

- breast cancer → BRCA1, BRCA2

+ Staging

① Size ② does it invade tissues or not.

③ lymph nodes, involved or not

④ metastases.

IVM staging
T₀ tumor size
(0, 1, 4)
↳ Bad

N = degree of lymph nodes

(0, 1, 2, 3)

No nodes
↳ Bad

M = metastases

(0, 1)

No → yes

Grade tumor

G(1, 4) (low differentiation)

• the higher the grade the higher diagnosis for patient "disease"